

MacBook Pro scream like a steam engine. So our first tool to explore here is known as bootloose, built around Docker containers (that look like VMs) by running *systemd* as PID 1 and *ssh daemon* to log into the container machines. The advantage is that there are not many dependencies except the Docker daemon and no need for any virtualisation support from hardware. bootloose can run both on GNU/Linux and MacOS, and even Dockerd can be run in the created container machines.

bootloose usage

Running bootloose is easy as there is no installation step; just grab the static binary and move to any location configured in your system path like */usr/local/bin*, etc. The following command downloads the latest version of bootloose, moves to the system path location, and makes that executable (requires *sudo* privileges on the GNU/Linux):

```
curl -sSLk "https://github.com/
k0sproject/bootloose/releases/
download/$(curl -sSLk https://github.
com/k0sproject/bootloose/releases
2>&1|grep releases|grep tag|awk '{print
$7}'|awk -F ' ' '{print $2}'|awk -F '/'
'{print $NF}'|head -1)/bootloose-linux-
amd64" -o /usr/local/bin/bootloose \
&& chmod +x /usr/local/bin/bootloose
```

If you don't have the *sudo* privileges on your GNU/Linux machine, the following command will download bootloose in the current directory and you can use it from there:

```
curl -sSLk "https://github.com/
k0sproject/bootloose/releases/
download/$(curl -sSLk https://github.
com/k0sproject/bootloose/releases
2>&1|grep releases|grep tag|awk '{print
$7}'|awk -F ' ' '{print $2}'|awk -F '/'
'{print $NF}'|head -1)/bootloose-linux-
amd64" -o ./bootloose \
&& chmod +x ./bootloose
```

Going forward, you can either prepend every bootloose command with *./* or add the current directory in your system path. Let's verify if the downloaded binary is all good and works by executing the *bootloose* command. You should see something like what is shown in Figure 1. This figure is a screenshot taken on my Ubuntu 18.04 LTS laptop.

The creation of container machines is driven by a configuration file, declaring various properties for the machines required by bootloose. Execute the command *bootloose config create* to create a configuration *bootloose.yaml* in your current directory. On my laptop, it looks like what's shown in Figure 2.

The basic configuration properties required by bootloose are self-explanatory. The *privateKey* is the ssh private key created by bootloose to let you ssh into the created container

machine(s). The *count* denotes the number of desired container machine(s) to be created. The created container machines are named as the combination of *cluster name*, *machines name* and *count* index starting from 0. The *image* is the Docker image already having *systemd* and *ssh daemons* to instantiate the container machine(s). The *portMappings* is about exposing your container port(s) on the localhost to access those. You can further configure the default values of the properties as desired through various flags passed to the *bootloose config create* command. Those flags are dumped using the *-h* or *--help* suffix to the command.

The *bootloose* project currently provides the prebuilt images of the container machines *quay.io/k0sproject/bootloose-{alpine3.18, alpine3.19, amazonlinux2023, amazonlinux2, clearlinux, debian10, debian12, fedora38, rockylinux9, ubuntu18.04,*

```
bootloose - Container Machines

Usage:
  bootloose [command]

Available Commands:
  completion  Generate the autocompletion script for the specified shell
  config      Manage cluster configuration
  create      Create a cluster
  delete      Delete a cluster
  help        Help about any command
  show        Show all running machines or a single machine with a given hostname.
  ssh         SSH into a machine
  start       Start cluster machines
  stop        Stop cluster machines
  version     Print bootloose version

Flags:
  -c, --config string  Cluster configuration file (default "bootloose.yaml")
  -h, --help            help for bootloose

Use "bootloose [command] --help" for more information about a command.
```

Figure 1: bootloose help

```
cluster:
  name: cluster
  privateKey: cluster-key
machines:
- count: 1
  spec:
    image: quay.io/k0sproject/bootloose-ubuntu20.04
    name: node%d
    portMappings:
    - containerPort: 22
```

Figure 2: bootloose default config created